

ZTA Solution Guide

The reference answers for every use case — pain-point responses, requirement responses, and expected products — reproduced from the official SME Solution Guide, with the customer architecture and each use case's reference solution diagram. **Proctor only — do not show participants.** Pair with the ZTA Grading Guide for the rubric, levels and scoring.

Overview

This Zero Trust design clinic was created to educate the field technical resources on how Cisco's Zero Trust architecture can help organizations strengthen enterprise security while preparing for the rise of an agentic workforce. Attendees will learn how to guide meaningful customer conversations, lead effective design discussions, and identify practical opportunities to apply Zero Trust concepts across the enterprise.

In the TechElevate Zero Trust Design Clinic, attendees will have heard how Zero Trust principals apply to the One Cisco architecture and will be given a chance to apply concepts and principles into real-world customer use cases.

Design Clinic Exercise

Welcome to this design clinic. In today's complex threat landscape, static network boundaries are no longer sufficient. To achieve a true Zero Trust posture, we must shift from reactive security to an iterative, dynamic approach.

This exercise challenges your group to design a Zero Trust environment to satisfy the following 5 use cases (3 foundational and 2 optional if time permits) using what you already know from Cisco and the new agentic AI topics taught in today's session. Work together as a group and each person should assume a role below to contribute towards your final proposal.

Roles in your group

Each person in your group will assume a role.

NetOps

- Responsible for network architecture and Operations
- Often includes desktop support

Information Security

- Responsible for information security and corporate security policy
- Responsible for IAM: AD, Entra, Sailpoint, etc.

Compliance

- Validate decisions made by the architects
- Validate policies align to customer requirements

Scribe

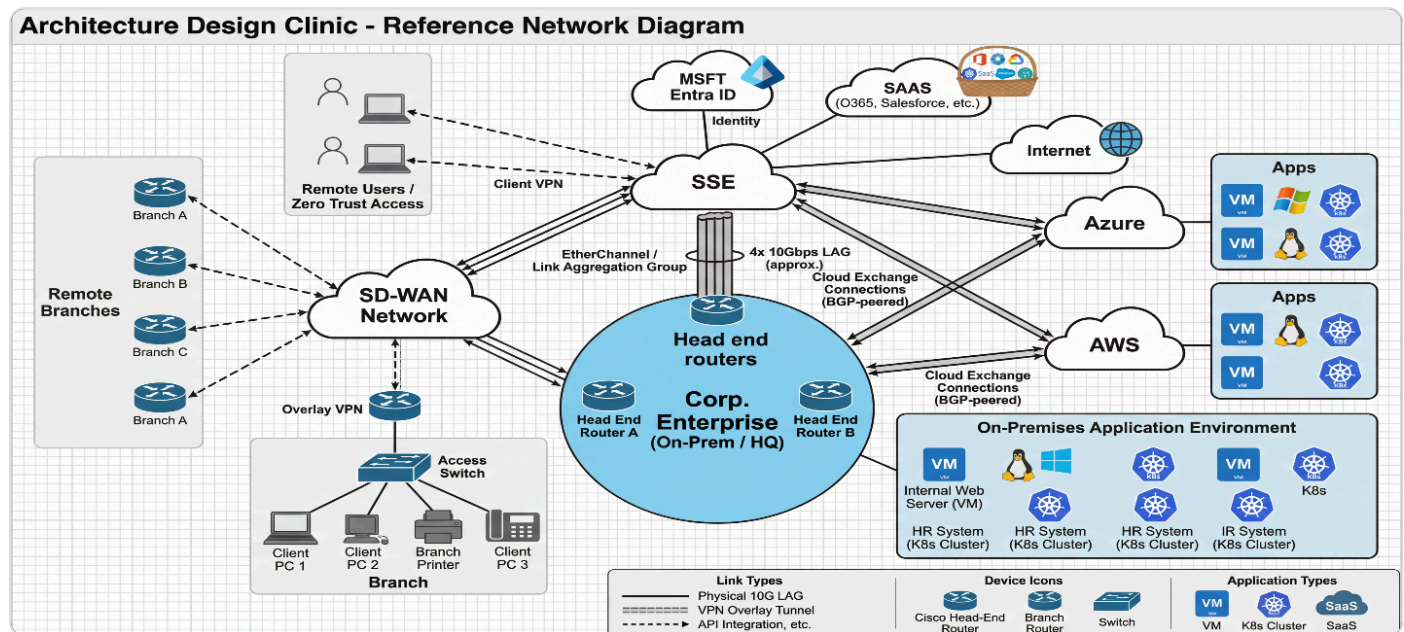
- Responsible for documentation
- Final diagram and responses

Current Customer Architecture Diagram

The customer is a distributed enterprise: a corporate HQ and on-premises data center, a fleet of remote branches on SD-WAN, a growing remote and hybrid workforce — some of them now running AI agents — and

an application estate spread across on-prem, AWS and Azure. They have grown by acquisition, they are mid-flight into public cloud, and their security controls have not kept pace with their footprint.

They are not asking you to rip anything out. They are asking you to bring Zero Trust to an environment that already exists — to segment it, to see it, and to keep enforcing policy no matter where the user, the device, or the workload happens to be.



Architecture Design Clinic — Reference Network Diagram (current customer environment)

Group Tasks

- Use AI tools (i.e. Cisco Circuit app) to help you draw out your final diagram. For example, upload a rough sketch and have AI produce a nice professional network diagram as part of your proposal.
- Be able to explain to your SME your proposed design for each use case listed below and why you designed it your way. List any Cisco advantages or differentiators if available.

Use Case #1 — Managed Environment; Segmentation

Foundation · Required

100 pts

CUSTOMER REQUIREMENTS

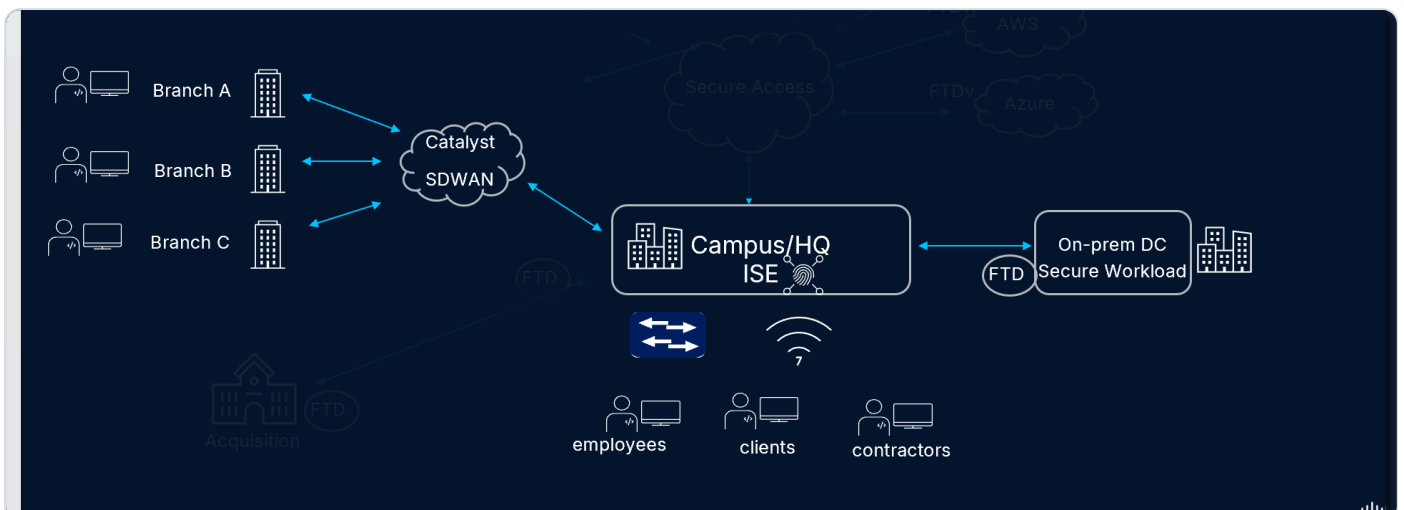
- Employee and trusted devices (Group=Employees)
- IOT (printers, cameras, badge readers, etc.) (Group=IOT)
- Deskside agents (dedicated mac-minis) (Group=deskagents)
- Guest SSID for Internet only (Group=Guest)
- IOT should be restricted from the other groups but need data center and Internet access
- Guests should have Internet access only
- Employee and IOT require controlled internet, and data center application access
- Deskside Mac-Minis have access to on-prem data center only

CUSTOMER PAIN POINTS

- Prior segmentation failed. Using VLANs/IP subnets is complex and tedious.
- Profiling is labor intensive and staff unable to keep up with constant new devices joining
- Failing PCI audits
- Corporate PCs not running latest AntiVirus and OS updates
- No budget for buying more firewalls to do segmentation

GROUP TASKS

- Draw a diagram of the customer's campus and branch wired/wireless networks. Submit for SME review and scoring.
- Create your proposed segmentation solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.



Use Case 1 — reference solution diagram (Segmentation: ISE + SGT across Campus/HQ, SD-WAN branches, on-prem DC Secure Workload)

Reference answer — addressing pain points

CUSTOMER PAIN POINT	CISCO TEAM RESPONSE
Prior segmentation failed. Using VLANS/IP subnets is complex and tedious.	Implement SGTs instead of VLAN/IP subnets
Profiling is labor intensive and staff unable to keep up with constant new devices joining	ISE cloud AI profiling
Failing PCI audits	Apply SGT for segmentation and insert FW services where L7 security services is required
Corporate PCs not running latest AntiVirus and OS updates	Use ISE Posture and remediate out-of-date PCs
No budget for buying more firewalls to do segmentation	Use SGTs to enforce segmentation within the wired/wireless network

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	CISCO TEAM RESPONSE
Employees / IOT / deskagents (Mac-minis) / Guest groups	Assign an SGT for each (Employees, IOT, deskagents, Guest). Authenticate employees using 802.1X or WebAuth; Mac-Minis can be profiled or added to the user's BYOD list or MAB; IOT can be profiled via MAB or manual SGT assignment
IOT should be restricted from the other groups but need data center and Internet access	Configure the SGT Matrix to only allow IOT to Data Center SGT and Internet. Deny for all other destination SGTs.
Guests should have Internet access only	Configure the SGT Matrix to only allow SGT=Guests to SGT=Internet.
Employee and IOT require controlled internet, and data center application access	Configure the SGT Matrix per source & destination SGTs
Deskside Mac-Minis have access to on-prem data center only	Configure the SGT Matrix: SGT=MacMini permit to SGT=on-premDC only.

Reference answer — expected products

PROPOSED PRODUCT / FEATURE	DESCRIPTION / HOW	WHY / DIFFERENTIATOR
ISE	Authenticate and profile every user and device (Web, 802.1X, MAB)	Visibility and scalable
ISE-SGT	Assign an SGT to each group (Employee, IOT, Desktop-Mini, Internet)	Cisco innovation; dynamic policy and topology-independent; end-to-end segmentation into multicloud Data Centers (on-prem and cloud providers)
ISE SGT ACL policy	Granular access policy	SGACL more efficient over IP ACLs
ISE pxGrid	pxGrid / Rapid Threat Containment; shares context with ecosystem devices	Cisco innovation, RFC8600; dynamically quarantines discovered threat devices; integrates with the Cisco portfolio (ACI, FW, SNA, SDWAN, CSW, SA, switches, routers, etc.)
Catalyst SDWAN	pxGrid integration for unified security policies via C3 / Security Cloud Control	—
FMC / SCC / FTD	Threat detection/protection via NGIPS, malware protection and content filtering	Integrates with ISE via pxGrid for Rapid Threat Containment
FMC / SCC / FTD — Source/Dest SGT policy enforcement	Dynamically blocks malicious hosts via SGT matching	Cisco innovation; dynamic policy enforcement
Cisco Secure Workload — App Dependency Mapping	Automatically profiles traffic flows and builds enforcement policy for application segmentation; ingests NetFlow from FTD for traffic profiling	Auto policy creation, validation and enforcement
Cisco Secure Workload — pxGrid integration with ISE	pxGrid / ISE Connector	Build granular segmentation policy based on identity and device posture learned from ISE/pxGrid

Use Case #2 — Remote Worker: Zero Trust Access

Foundation · Required

100 pts

CUSTOMER REQUIREMENTS

- All corporate users can use full client VPN to access selected legacy private applications. But most prefer ZTNA for modern applications.
- All corporate users need controlled internet access (block gambling and gaming)
- Contractors can only access private applications via clientless RDP access.
- Improve user experience with minimal application sign-ons
- Customer wants SASE solution for branches

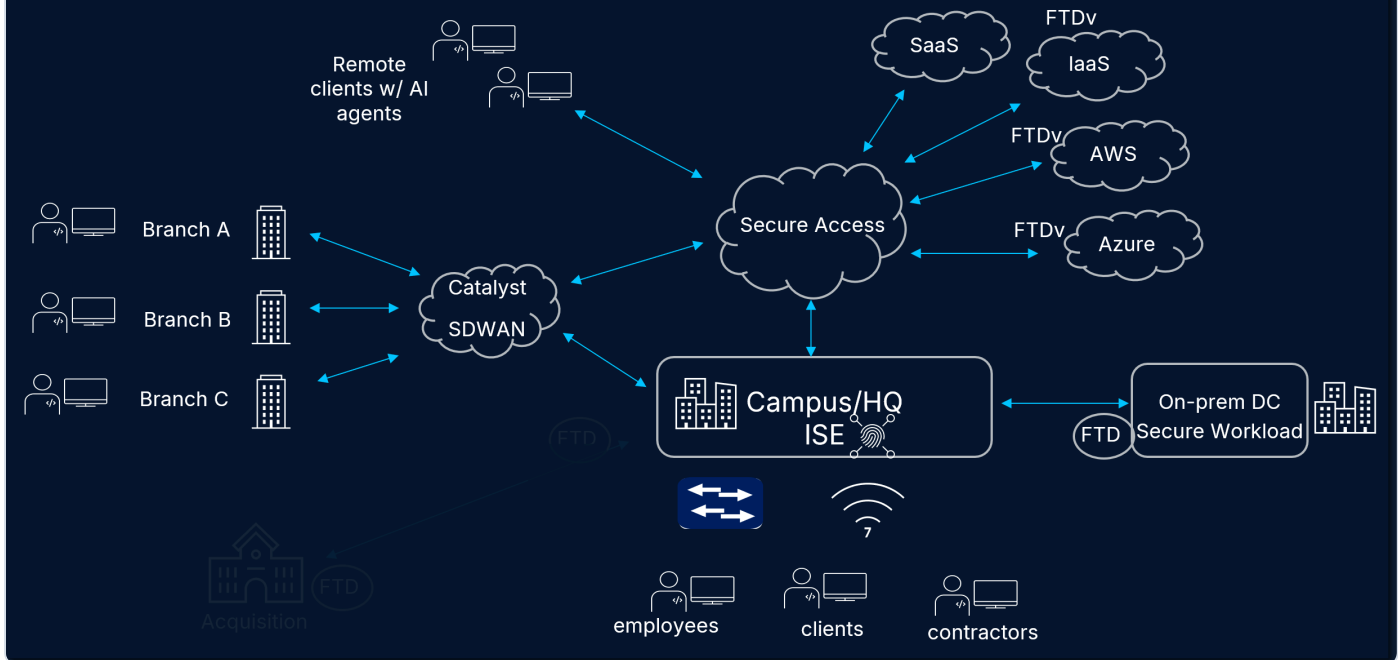
CUSTOMER PAIN POINTS

- Full client VPN has over permissive privileges. Not considered Zero Trust
- User satisfaction is low due to manual VPN connection daily
- Too many sign-ons per application. Low user satisfaction.
- Want to eliminate password logins to minimize stolen credentials
- Existing Okta IdP solution getting too expensive
- CIO wants SASE to reduce branch costs

GROUP TASKS

- Expand the diagram of the customer's campus and branch wired/wireless networks to support remote workers. Submit for SME review and scoring.
- Create your proposed secure remote worker solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.

Use Case #2: proposed diagram for SSE



Use Case 2 — reference solution diagram (SSE / Secure Access for remote workers, SaaS/IaaS/AWS/Azure via FTDv, SD-WAN branches)

Reference answer — addressing pain points

CUSTOMER PAIN POINT	CISCO TEAM RESPONSE
Full client VPN has over permissive privileges. Not considered Zero Trust	Use Zero Trust Access from Secure Client to connect to Secure Access
User satisfaction is low due to manual VPN connection daily	Use the Zero Trust Access module to automatically deliver a per-application VPN tunnel to corporate applications
Too many sign-ons per application. Low user satisfaction.	Use Duo Passport for SSO
Want to eliminate password logins to minimize stolen credentials	Use Duo advanced passwordless
Existing Okta IdP solution getting too expensive	Use Duo Directory
CIO wants to modernize branches with SASE	Integrate SDWAN into the Secure Access cloud with Direct Internet Access

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	CISCO TEAM RESPONSE
All corporate users can use full client VPN to access selected legacy private applications. But most prefer ZTNA for modern applications.	Use the ZT module in Secure Client and create private resources with only ZT-module access; all other corporate apps can be configured with full VPN client access
All corporate users need controlled internet access (block gambling and gaming)	Use Secure Web Gateway to block categories gambling and gaming for corporate employees
Contractors can only access private applications via clientless RDP access.	Configure selected private resources to be accessible via browser (TCP) only for contractor groups
Improve user experience with minimal application sign-ons	Use Duo Passport or SSO
Customer wants to modernize branches with SASE	Cisco Secure Access easily integrates with Cisco Catalyst SDWAN to modernize Layer-7 services from the cloud

Reference answer — expected products

PROPOSED PRODUCT / FEATURE	DESCRIPTION / HOW	WHY / DIFFERENTIATOR
Secure Access, Secure Client, Duo — ZTA module in Secure Client	The ZT Access module can deliver per-app VPN access to private resources	Ease of use; no full VPN required; meets Zero Trust least-privilege access requirement
Secure Client VPN module (Full Tunnel)	VPN module can provide access to legacy applications	Mature VPN technology; assigned with an SGT for segmentation
ZTA module, Secure Client	Intercepts all ports/protocols traffic from the client and redirects traffic to Secure Access for inspection	Control internet-bound traffic with SWG policy; block gambling and gaming
Secure Access Private Resources proxy	Proxy for private resources; can publish private apps for VPN client, ZTA module, or browser only (guests, contractors — SSH or RDP)	Easy provisioning of private apps for various users (VPN, ZTA module, or browser)
Duo	Duo Passport w/ SSO	OS SSO login is unique and provides one login experience
Secure Access VPNaaS	VPNaaS over on-premises VPN hardware at HQ	VPN traffic can map to an SGT for segmentation — unique to Cisco
Secure Access Unified Policy	Unified policy for public and private resources; integrated with SCC/CCC	Unique to Cisco; ease of use
Secure Access + SDWAN	SASE integration with SDWAN via IPsec tunnels	Ease of deployment from SDWAN to Secure Access

Use Case #3 — AI Access and Agent Controls

Foundation · Required

100 pts

CUSTOMER REQUIREMENTS

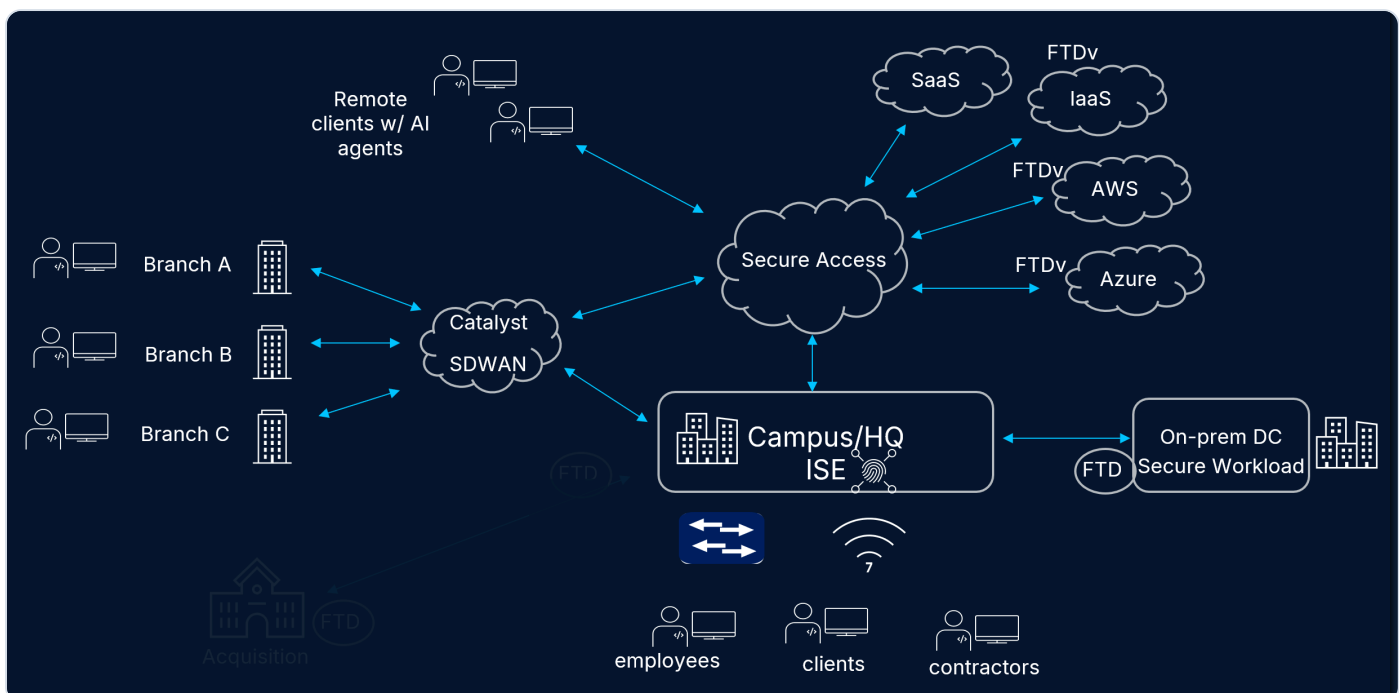
- User agents should not be granted rights to use tools equally. Fine grained control is needed.
- Corporate users should only be able to login to their organization's ChatGPT tenant
- Some remote users will be running Claude Desktop agents. Ensure your design includes controls for those employee systems.
- Some users will be integrating MCP servers to local agents like Claude Code

CUSTOMER PAIN POINTS

- Shadow AI usage is an issue today
- Too many unknown MCP servers for applications in both the private and public DCs.
- Users using personal credentials for non-enterprise ChatGPT login
- Downloading of risky AI models

GROUP TASKS

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed agentic solution based on on what was taught earlier in the agentic AI design presentation and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.



Use Case 3 — reference solution diagram (Secure Access AI Access / AI Gateway for remote clients with AI agents)

Reference answer — addressing pain points

CUSTOMER PAIN POINT	CISCO TEAM RESPONSE
Shadow AI usage is an issue today	Use Secure Access's built-in AI Access feature to begin control of corporate AI usage.
Over-privileged MCP servers for applications in the private and public DCs	Use Secure Access's AI Gateway solution to register and manage allowed MCP servers
Users using personal credentials for ChatGPT login	Use Secure Access's tenant controls for AI
Downloading of risky AI models	Secure Access's semantic inspection will identify and block malicious AI models

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	CISCO TEAM RESPONSE
Users should not be granted rights to use tools equally. Fine grained control is needed	Duo / Secure Access MCP integration can provide fine-grained AI-tools control along with authentication
Corporate users should only be able to login to their organization's ChatGPT tenant	Secure Access tenant control can provide this feature for ChatGPT
Some remote users will be running Claude Desktop agents. Ensure your design includes controls for those employee systems.	Duo IdP provides the agent registration process (DCR) to manage and control what registered desktop agents can do
Some users will be integrating MCP servers to local agents like Claude Code	Register and govern MCP servers through the Secure Access AI Gateway with Duo authorization

Reference answer — expected products

PROPOSED PRODUCT / FEATURE	DESCRIPTION / HOW	WHY / DIFFERENTIATOR
AI Access within Secure Access	AI control with guardrails	Cisco is helping organizations bridge this gap by integrating the advanced data loss prevention (DLP) capabilities of Cisco Secure Access with the specialized protection of Cisco AI Defense, which provides a comprehensive security framework that monitors AI application usage, enforces guardrails, and protects your data across endpoints, cloud, and web environments.
AI Gateway within Secure Access	AI Gateway controlling registered MCP servers	—
Duo IdP	Duo provides Fine Grained Authorization and Agentic registration (DCR)	Duo has the easiest integration with an AI Gateway such as that of Secure Access, as well as the flexibility to integrate with 3rd-party IdPs where most IdPs are focused only on the latter

Use Case #4 — Merger / Acquisition; Extend Segmentation

Bonus · If time permits

100 pts

CUSTOMER REQUIREMENTS

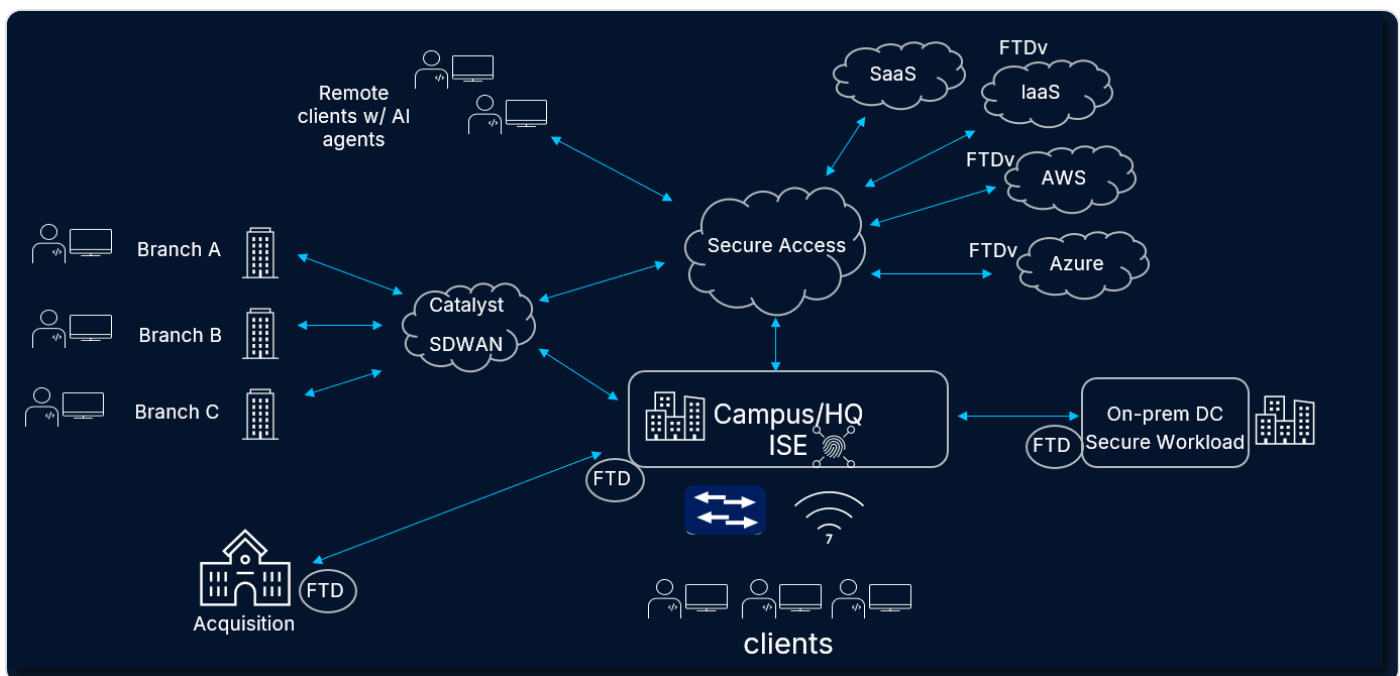
- CISO wants to extend segmentation to acquired company using TrustSec.
- Only selected branch users can access a restricted resource at recently acquired company (Site 9951).

CUSTOMER PAIN POINTS

- Parent company is not ready to add 100 newly acquired staff into their corporate directory. Is there a quick option of using what they already have?
- During transition, minimize disruptive user authentication experience to corporate apps and quickly secure acquired company apps/user population
- Acquired company suffered identity breach previously. Still using passwords.

GROUP TASKS

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.



Use Case 4 — reference solution diagram (extend TrustSec to the acquired site via FTD inline SGT tagging; Duo Directory)

Reference answer — addressing pain points

CUSTOMER PAIN POINT	CISCO TEAM RESPONSE
Unsure how to extend segmentation to acquired company	Use SGT capability in the remote FTD and enable inline tagging to extend SGT tags from HQ to the remote company.
Parent company not ready to add 100 newly acquired staff into their corporate directory. Is there a quick option of using what they already have?	Use Duo Directory to host the 100 acquired staff. Duo can import users from Okta via SCIM
During transition, minimize disruptive user authentication experience to corporate apps and quickly secure acquired company apps	Use Duo SSO routing rules to provide a seamless authentication experience. Apps protected by Okta can still authenticate with Okta until the customer is ready to migrate off Okta.
Acquired company suffered identity breach previously. Still using passwords.	Use Duo Passwordless

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	CISCO TEAM RESPONSE
CISO wants to extend segmentation to acquired company using TrustSec	Use SGT capability in the remote FTD and enable inline tagging to extend SGT tags from HQ to the remote company.
Only selected branch users can access a restricted resource at recently acquired company (Site 9951)	Can use source & destination SGTs in the remote FTD access-control policy

Reference answer — expected products

PROPOSED PRODUCT / FEATURE	DESCRIPTION / HOW	WHY / DIFFERENTIATOR
FMC / FTD	FTD to build a Site-to-Site VPN tunnel from the remote company to the HQ FTD. FTD can do inline SGT support per interface.	Cisco SGT inline tagging is a differentiator over others
Duo — IdP (Directory)	Duo Directory; use SCIM to import users from Okta easily.	Duo ease of use and integration with others to bring overall value
Duo — IdP (SSO routing)	Duo SSO routing rule: maintain a consistent user-auth experience from old Okta to new.	Duo ease of use and integration with other IdPs to bring overall value
SCC	SCC to configure SGT policy into SDWAN. Match source SGT (branch) and destination SGT (acquisition FTD)	SCC provides unified SGT policy management.

Use Case #5 — Zero Trust across multicloud Data Centers

Bonus · If time permits

100 pts

CUSTOMER REQUIREMENTS

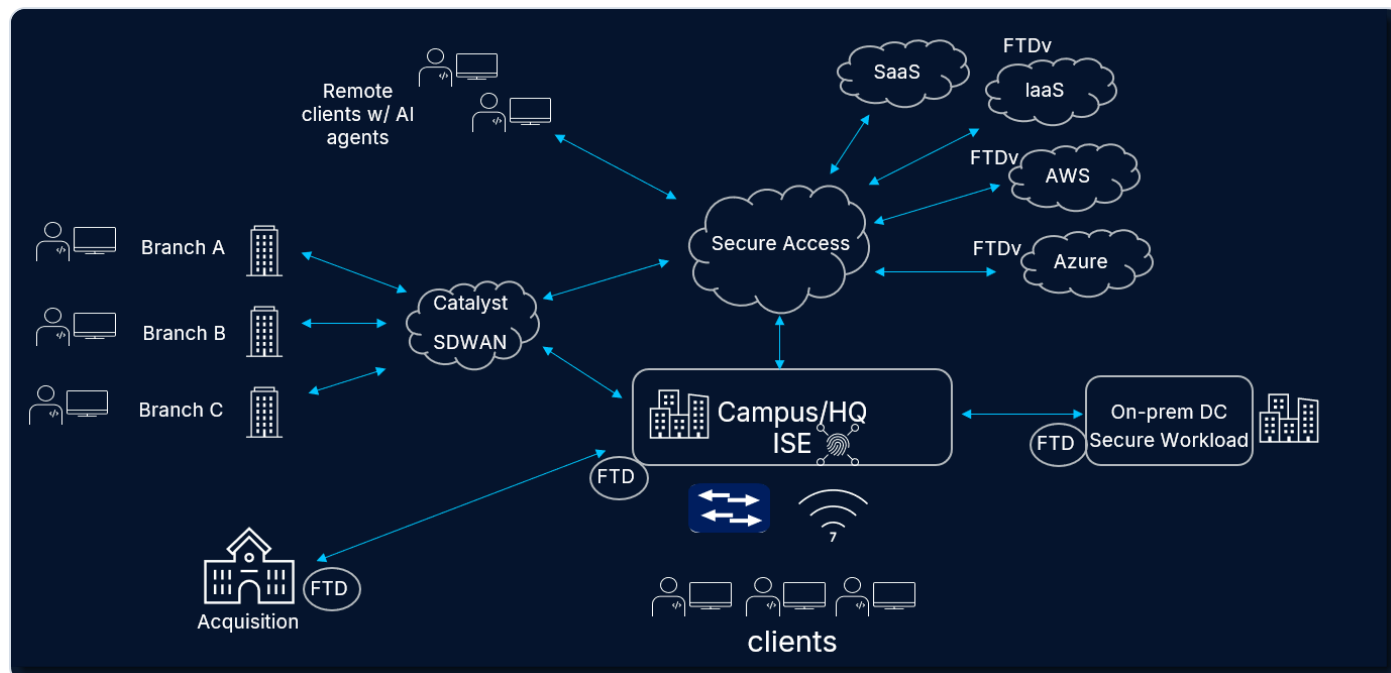
- Limited budget. Want to leverage native security features when practical. Open to virtual appliances
- CIO seeking end-to-end segmentation solution
- Minimize complexity

CUSTOMER PAIN POINTS

- Minimal visibility of applications across multicloud data centers
- Lacking segmentation strategy and plan
- Wide deployment of Kubernetes with latency and visibility issues
- Unable to patch IOT devices
- Too many disparate security management interfaces for on-prem and multicloud enforcement points

GROUP TASKS

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.



Use Case 5 — reference solution diagram (Secure Workload discovery & microsegmentation across on-prem, AWS and Azure)

Reference answer — addressing pain points

△ **Source note.** In the source Solution Guide this “addressing pain points” table repeats Use Case 4’s answers (a copy-paste error) and is reproduced verbatim here. When grading, map the UC5 pain points to the requirement and product responses below (Secure Workload App Dependency Mapping for visibility; auto-baseline policy for the segmentation

plan; Isovalent/Cilium/Tetragon for Kubernetes; Secure Workload/FTD + runtime protection for unpatchable IOT; Cisco Cloud Control to unify consoles).

CUSTOMER PAIN POINT	CISCO TEAM RESPONSE
Minimal visibility of applications across multicloud data centers	Use inline SGT tagging in FTD device connecting to parent company FTD. Can assign SGT to all acquired staff
Lacking segmentation strategy and plan	Use Duo Directory to host 100 acquired staff. Duo can import users from Okta via SCIM
Wide deployment of Kubernetes with latency and visibility issues	Use Duo SSO routing rules to provide seamless authentication experience. Apps protected by Okta can still authenticate with Okta until the customer is ready to migrate off Okta.
Unable to patch IOT devices	Use Duo Passwordless
Too many disparate security management interfaces for on-prem and multicloud enforcement points	Cisco Cloud Control unifies multiple UIs into one to simplify policy creation and deployment

Reference answer — meeting requirements

CUSTOMER REQUIREMENT	CISCO TEAM RESPONSE
Limited budget. Want to leverage native security features when practical. Open to virtual appliances	Use Cisco Secure Workload to help discover and manage workloads and apps. CSW can integrate with AWS, Azure and others on access-control policies; if the customer is open to cloud virtual appliances, CSW can also sync access policies between CSW and FTDv.
CIO seeking end-to-end segmentation solution	ISE common policy can share context via pxGrid to CSW and FTD. ISE can build an end-to-end segmentation policy engine
Minimize complexity	ISE common policy and CSW discovery can unify DC firewalling and policy control.

Reference answer — expected products

PROPOSED PRODUCT / FEATURE	DESCRIPTION / HOW	WHY / DIFFERENTIATOR
Secure Workload	Agent and agentless design; App Dependency Mapping (discovery); auto-baseline policy creation; granular host policy enforcement; SGT integration with ISE; cloud (AWS, Azure) platform integration	CSW provides comprehensive end-to-end policy
NGFW / FTD	FTD at the edge for agentless networks; flow info from FTD to CSW profiles the traffic baseline; CSW-to-FTD auto-policy deployment	Cisco's CSW integration with FTD provides the best end-to-end solution
Isovalent	eBPF for kernel-level protection; Cilium CNI for K8s protection (observability, security, LB, BGP); Tetragon for runtime protection	Isovalent's advantage of kernel-level protection via eBPF